SigLIP · arXiv 2303.15343 · Sep 14

Scientific Reviewer

seat 5 · 8+3

Strengths · weaknesses · actionable fixes

Paper club · VLA Architectures · dark academic

Strengths

Clean loss + systems co-design

Sigmoid + chunked distributed impl — not just a loss paper

Rare mega-batch study

Sweeps 512 → 1M; honest saturation / harm past ~32k

Actionable ablations

Bias b, β_2, WD-on-unlock — reimplementable knobs

Public So400M checkpoint

Efficiency point that actually shipped into open VLAs

Locked + unlocked recipes

SigLiT vs SigLIP — both useful for different budgets

Weaknesses

Proprietary WebLI

Absolute numbers hard to audit; public-data gap

Softmax baseline systems

CLIP-style baseline less systems-optimized than SigLIP path

Thin theory

Mostly empirical pairwise logistic motivation

Limited dense / VLM-as-encoder eval

At publication — later PaliGemma fills some of this

Compute-matched SOTA imperfect

Hardware-generation mixes vs CLIP/FLIP lore

Short captions (max 16 tok)

May under-serve long VLA instructions unless FT compensates

Actionable fixes

Gift list — no motivation tour

1. **Public-data replication**
   CC/COCO/Flickr or open WebLI subsets; report small-BS gap
2. **VLM / robotics transfer**
   Report encoder quality on robot stills / instruction retrieval
3. **Publish So400M→PaliGemma config**
   Exact freeze/unlock, resolution, token length
4. **Isolate loss vs data**
   When claiming “SigLIP helps VLA,” ablate loss separately
5. **Long-caption stress**
   Eval >16-token instructions; NaFlex / resolution for egocentric
6. **With/without bias replications**
   Secondary lit often misses $b \approx -10$ — make it first-class

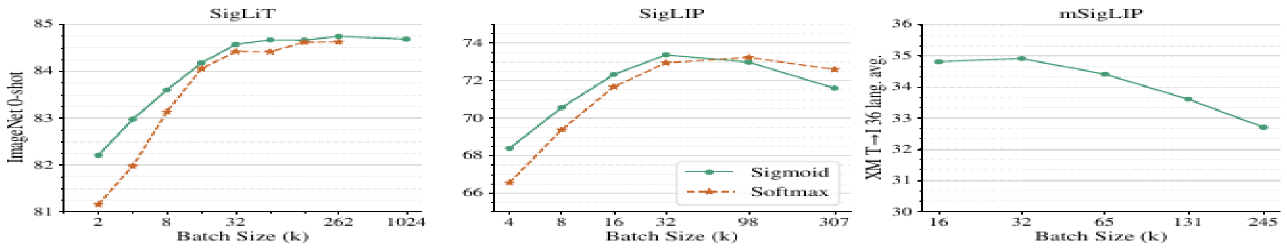

Figure 2: The effect of pre-training batch size. **Left: SigLiT results**, trained for 18B seen examples. Sigmoid loss outperforms the softmax loss significantly with small batch sizes, and performs similarly at larger batch sizes. We successfully trained an SigLiT model with up to *one million* batch size. However, performance for both sigmoid and softmax saturate at around 32 k batch size. **Middle: SigLIP results**, trained for 9B seen examples. Both sigmoid loss and softmax loss saturate at a reasonable batch size, while the peak of the sigmoid loss comes earlier and slightly outperforms the peak of the softmax loss. A very large batch size hurts both losses. **Right: mSigLIP results**, trained for 30B seen examples. With a multilingual setup using over 100 languages, 32 k batch size is surprisingly sufficient and scaling beyond that hurts performance on a 36-language cross-modal retrieval task.

method. In words, we first compute the component of the loss corresponding to the positive pairs, and $b-1$ negative pairs. We then permute representations across devices, so each device takes negatives from its neighbouring device (next iteration of sum **B**). The loss is then calculated with respect to this chunk (sum **C**). This is done independently in each device, such that each device computes the loss with respect to its local batch b. Losses can then simply be summed across all devices (sum **A**). Individual collective permutes (for sum **B**) are fast (and indeed D collective permutes is typically faster than two all-gathers between D devices), and the memory cost at any given moment is reduced from $|\mathcal{B}|^2$ to b^2 (for sum **C**). Usually b is constant as scaling $|\mathcal{B}|$ is achieved by increasing the number of accelerators. Due to being quadratic with respect to the batch size, the vanilla loss computation rapidly bottlenecks scaling up. This chunked approach enabled training with batch sizes over 1 million on relatively few devices.

4. Results

In this section, we evaluate the proposed SigLiT and SigLIP models across a wide range of batch sizes. We discuss what can be achieved with a small number of accelerator chips, using both SigLiT and SigLIP recipes. We also briefly discuss the impact of batch size on multilingual language image pre-training. We ablate the importance of our large-batch stabilization modification and the introduced learned bias term and present a study on the effect of positive and negative pairs ratio in the sigmoid loss. Lastly, we explore SigLIP's data noise robustness.

To validate our models, we report zero-shot transfer results on the ImageNet dataset [14] and zero-shot retrieval results across 36 languages on the XM3600 dataset [44]. We use the ScalingViT-Adafactor optimizer [58] by default for all our experiments.

4.1. SigLiT: Scaling batch size to the limit

Following [59], we use the same precomputed embeddings for the images using a ViT-g vision model, and train a base size text tower from scratch with the same hyperparameters using the LiT image-text dataset [59].

We perform a study over a wide range of batch sizes, from 512 to 1 M, demonstrating the impact of batch size for contrastive learning. Results are presented in Figure 2 (left). When the batch size is smaller than 16 k, sigmoid loss outperforms softmax loss by a large margin. With growing batch sizes, we observe that softmax loss quickly catches up and potentially slightly underperforms sigmoid loss with a large enough batch size. Overall, we recommend using the SigLIP recipe for large batch sizes as well, due to the simplicity, compute savings, and straightforward memory efficient implementation.

There is a consensus that contrastive learning benefits from large batch sizes, while most of the existing studies stop at 64 k batch size [59, 35, 10]. We successfully trained an SigLiT model at one million batch size, to explore the limit of contrastive learning. To our surprise, the performance saturates at 32 k batch size, further scaling up the batch size only gives a minor boost, and the model peaks at

Evidence to cite when reviewing the batch narrative

Extra critique notes

Negatives · bias under-sold · downstream ≠ loss · locked-vision speed

Reviewer close

Strengths real · WebLI gap · isolate loss vs data